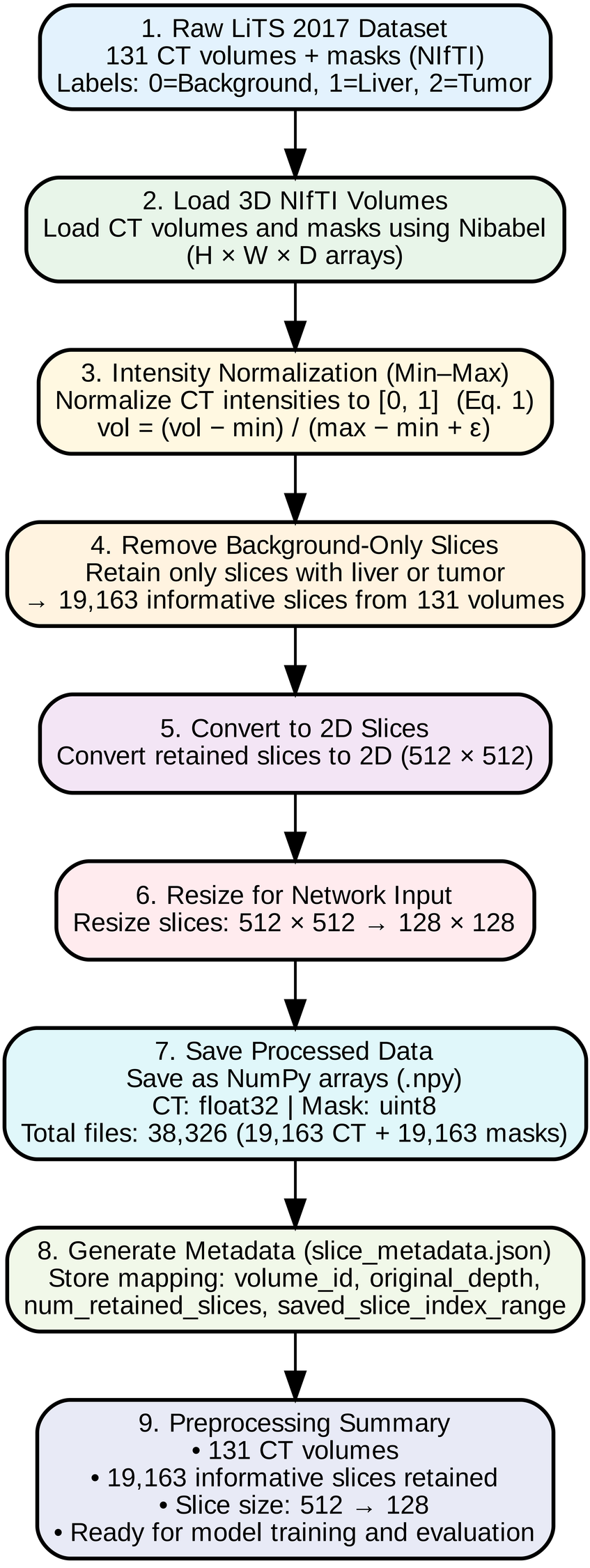
1. Raw LiTS 2017 Dataset
131 CT volumes + masks (NIfTI)
Labels: 0=Background, 1=Liver, 2=Tumor
2. Load 3D NIfTI Volumes
Load CT volumes and masks using Nibabel
(H × W × D arrays)
3. Intensity Normalization (Min–Max)
Normalize CT intensities to [0, 1] (Eq. 1)
vol = (vol − min) / (max − min + ε)
4. Remove Background-Only Slices
Retain only slices with liver or tumor
→ 19,163 informative slices from 131 volumes
5. Convert to 2D Slices
Convert retained slices to 2D (512 × 512)
6. Resize for Network Input
Resize slices: 512 × 512 → 128 × 128
7. Save Processed Data
Save as NumPy arrays (.npy)
CT: float32 | Mask: uint8
Total files: 38,326 (19,163 CT + 19,163 masks)
8. Generate Metadata (slice_metadata.json)
Store mapping: volume_id, original_depth,
num_retained_slices, saved_slice_index_range
9. Preprocessing Summary
• 131 CT volumes
• 19,163 informative slices retained
• Slice size: 512 → 128
• Ready for model training and evaluation